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Innovative Approach of Generative AI for Automating Technical Bid Evaluations in Oil Companies

O. E. Abdelaziem and A. Nasser Khafagy, Amal Petroleum Company, Egypt; T. A. Yehia, Texas A&M University, USA

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Summary

The process of outlining a scope of work and evaluating technical bids in the oil and gas industry is commonly burdensome, labor-intensive, and susceptible to human bias. This paper introduces an AI-assistant chatbot based on the power of open-source large language models (LLMs), natural language processing (NLP), and data analytics, to aid in automating the entire workflow of technical tendering processes, facilitating an improved decision support system (DSS), and mitigating potential subjectivity.

The workflow starts with loading documents in the format of scanned PDF files. Firstly, data was extracted using optical character recognition (OCR), and text mining techniques. Secondly, Langchain was implemented to optimally split the document into chunks with adequate overlapping. Thirdly, embeddings were created using sentence transformers, and a vector store was constructed. After that, LlaMa model, was optimized and employed to query the vector store efficiently. Then, retrieval augmented generative (RAG) query engines were used to retrieve the matching documents and generate the final answer. Finally, a chat memory buffer was incorporated to maintain context and initialize a chatbot.

Implementing LLM in the tendering process has proven promising for reviewing and ranking the technical data of the submitted bids based on pre-defined criteria. However, it was shown that extracting tabulated data effectively from scanned documents is challenging. Consequently, it was essential to pre-process documents and extract tables into structured databases before querying them. In addition, model augmentation was employed to avoid hallucination and enhance the model reasoning and capability of identifying pass and fail criteria, nonetheless, human feedback is required to assess satisfying performance. Recall-Oriented Understudy for Gisting Evaluation (ROUGE) was found valuable to expedite the comparison between the submitted data in technical tables and the detailed data sheets. Furthermore, LLM was found effective in accelerating the process of market survey comparison, by retrieving and comparing prices and conditions of different documents. Moreover, supervised fine-tuning (SFT) was implemented to improve the model's capacity to compare clauses of the scope of work along with terms and conditions across different contracts, in order to discern major inherent differences and reduce potential bias in the future. This enhancement was configured through quantized low-rank adaptation (QLoRA) and parameter-efficient fine-tuning (PEFT).

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This paper presented a novel approach, which integrated LLMs into procurement information development systems, automating multiple tasks in the tendering process in the oil and gas upstream industry. This is the first time that such an application has been applied to the oil industry, which shall provide a foundation for future research.

Introduction

In the oil and gas industry, the evaluation of technical bids is a multidisciplinary task that requires collaboration between engineering and non-technical teams within a bid evaluation committee. This task is intricate, labor-intensive, and often prone to human bias and inefficiencies. It typically involves thorough analysis of technical and non-technical documents to ensure that awards align with operational requirements. Despite challenges persist, particularly in terms of timely and accurate decision making. Although recent advancements in AI, the applications of data analytics, OCR, and LLMs have been examined on various applications that could assist with bids evaluation in different fields, their application in oil and gas sector remains unexplored.

This paper introduces a novel workflow that integrates LLMs along with data analytics and OCR to enhance the technical evaluation of bids in oil integrated companies. This study focuses exclusively on the technical aspects of bid evaluation only, and did not address the entire bid documents. It primarily addresses 1) the data transformation from scanned documents into readable formats using OCR-based parsers, and 2) the implementation of open-source LLM for data retrieval and generation.

The findings from this study are expected to offer valuable insights for researchers aiming to harness AI to enhance decision support systems and drive innovation in technical evaluations of bids in the oil industry.

This paper is structured as follows: *Introduction* section provides an overview of the paper objective and depicts the problem statement. *The literature review* section examines the relevant preceding studies. *Methodology* section details the workflow employed in this study. *Results* are presented, followed by a discussion of findings. *Challenges and future work* outline the basis for further research and potential contributions. *Conclusion* offers a summary of the work done.

Problem Statement. Despite recent progress in AI, oil companies continue to rely on traditional approaches for processing and evaluating procurement and bids, which involves tedious manual reviews. It was noted that the implementation of LLMs solely, without being integrated with data analytics and ML techniques, resulted in unsatisfactory outcomes and significant misinterpretations due to the complex, unstructured nature of bid documents, which are typically in the form of scanned documents. Moreover, previous studies implemented LLMs agents, which are connected to the internet, however, oil companies are in favor of on-premises solutions that could utilize open-source LLMs, that could be run locally, due to privacy concerns.

The first section of this work, which addressed text extraction from scanned documents, was inspired by Dong et al. (2024), who developed and optimized an OCR system that could reasonably extract data from bid documents of different formats, thereby reducing the time required for review and verification as well as minimizing potential errors. The second section of this work, which addressed LLM workflow for data retrieval and generation, was inspired by the work done by Waseem et al. (2024), which proposed an AI procurement assistant that incorporates LLM agents to facilitate bid evaluations.

Literature Review

The integration of LLMs with DSS has recently emerged as it could be adapted to diverse project demands. This section provides a brief overview of LLMs, reviews recent advancements in LLMs advancements in oil industry within the context of text retrieval and examines LLMs use cases in bids and procurement processes.

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The first successfully trained LLM was introduced in year 2018 after being exposed to a vast amount of text to learn from (Pinheiro et. al 2023). An LLM is characterized by its large number of parameters that undergoes pretraining tasks such as masked language modeling, to understand human language by capturing semantics and modeling probabilities from extensive amount of text data. LLMs feature a profound understanding of natural language context, the ability to generate human-like text, contextual awareness, and strong instruction-following ability, all of which are valuable for problem-solving and decision-making (Yao et al. 2024).

Transformers. Fundamentally, LLMs were built on transformer neural network architecture, introduced by Vaswani et al. (2017). The core idea behind transformers is the implementation of attention mechanisms, where the representation of any position in the input is calculated based on a weighted combination of other positions, resulting in a complex system. Transformers consist of encoders, which tokenize the input and create an embedding vector representation. Self-attention score could be calculated based on embeddings and three key metrics – query, value, and key – which measures how much focus to place on different words in a sentence. Multiple self-attention heads are combined into a multi-head attention, constructing the foundation of transformers (Vaswani et al. 2017).

Review of LLMs in Oil Industry. LLMs have proven to be beneficial in enhancing DSS and reducing potential human bias across different fields, including the oil and gas industry, with different tasks such as well construction planning and real-time operations in the oilfield (He et al. 2021; Ogundare et al. 2023; Pacis et al. 2024). LLMs showed the power of delivering personalized, context-aware responses when dealing with databases of drilling and production PDF reports (Singh et al. 2023). In addition, LLMs showcased their ability in the retrieval and analysis of unstructured data, providing valuable insights rapidly (Martin et al. 2024a). Moreover, Martin et al. (2024b) presented a natural language querying system that could effectively search a large repository of unstructured exploration datasets and automatically summarize contexts.

A summary of the key research on the applications of LLMs in oil industry is depicted in (Table 1). These studies acknowledged potential challenges and limitations accompanied with the implementation of LLMs, such as data privacy and misleading information.

Table 1—A summary of recent studies on LLMs applications in the oil and ga industry

Reference	Application	LLM/AI Techniques Used	Key Findings/Benefits	Challenges Addressed
(Pacis et al. 2024)	Zero-shot learning for drilling information retrieval	GPT-4, zero-shot learning	Effective information retrieval without domain-specific training	Potential for hallucination, need for continuous updates
(Martin et al. 2024b)	Natural Language querying of upstream datasets	LLMs, embeddings, indexing	Fast, accurate search results, improved data discovery	Handling unstructured data, accurate indexing
(A. Dubovik et al. 2024)	Geophysical assistance, document analysis, business correspondence	Transformer architecture, open-source LLMs	Improved decision-making, enhanced document analysis	Resource-intensive, data privacy
(Tharayil et al. 2024)	Enhancing Knowledge Management Systems (KMS)	NLP, DL, LLMs, FAISS indexing	Efficient document extraction, improved search capabilities	Handling heterogeneous data, accuracy in retrieval
(Singh, Jia, and Nalagatla 2023)	Drilling and production analytics	GPT, cloud-native AI	Real-time data analysis, operational optimization	Potential for incorrect answers, data curation needs
(Yi et al. 2024)	Well construction planning and real-time operation	GPT, RAG framework	Enhanced information retrieval, context-aware responses	Risk of incorrect answers, data quality concerns

Applications of LLMs in Procurements. Data-driven approaches hold the potential to be implemented in procurement DSS in order to ensure that the submitted proposals by bidders meet the required criteria.

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Mandale and Kasamani (2022) introduced a tree-based model for effective tender evaluation, addressing inconsistencies and lack of objectivity associated with manual criteria in the evaluation process. Their model provides a pass or fail result for each bidder. Mirachi et al. (2022) developed a framework for DSS in tendering process, demonstrating its potential to save efforts required from multidisciplinary teams. They highlighted the importance of LLMs, noting that while digital systems have already improved the databases of documents, they have not fully ensured the correct exploitation of information stored in them. Waseem et al. (2024) introduced an AI-procurement assistant that incorporates LLM agents to improve intelligent search and bids evaluation. They assessed bids based on quoted prices, services quality, supplier history, commitment, and additional benefits. Their ultimate objective was to automatically select an outstanding proposal that aligns with assessment standards. They emphasized the necessity to automate bidding evaluation process due to the extensive manual efforts in traditional evaluation along with inherent risk of human-related errors.

In the oil industry, Faraji et al. (2022) developed a bidding decision support model that leverages historical data along with expert judgment using a dataset of more than 100 contracts in the downstream sector. The model achieved credibility of 75% in both cost and time evaluations.

Methodology

This workflow addressed the resolution of two key challenges that technical engineering teams encounter when working typically with procurement teams on bids. The first pertains to benchmark studies and market scans, while the latter is centered on the assessment of technical datasheets submitted by bidders against the company's required specifications. This solution shall accelerate the evaluation phase and eliminate redundant work.

A systematic workflow depicted in (Fig. 1) was meticulously designed to ensure viable outcomes. Initially, Llamindex parser and OCR were used to transform scanned documents into structured, readable formats. Next, the approach of human-in-the-loop was implemented, where the extracted data from the previous step are reviewed and refined as needed – this step is crucial to facilitate the effectiveness of the subsequent processes. Then, the open-source LlaMa model was employed locally on computers to handle complex queries and chaining tasks. Following this, the integration between llamaindex and langchain enhanced the performance of question-answering and summarization tasks. Subsequently, embedding was established to construct a vector database, simplifying text similarity and grouping. Next, RAGs were implemented to prevent the model from reasoning beyond the provided context. Ultimately, chat-buffering maintained coherence and contextually relevant responses in prolonged chat conversations. Further elaboration of each step is illustrated below:

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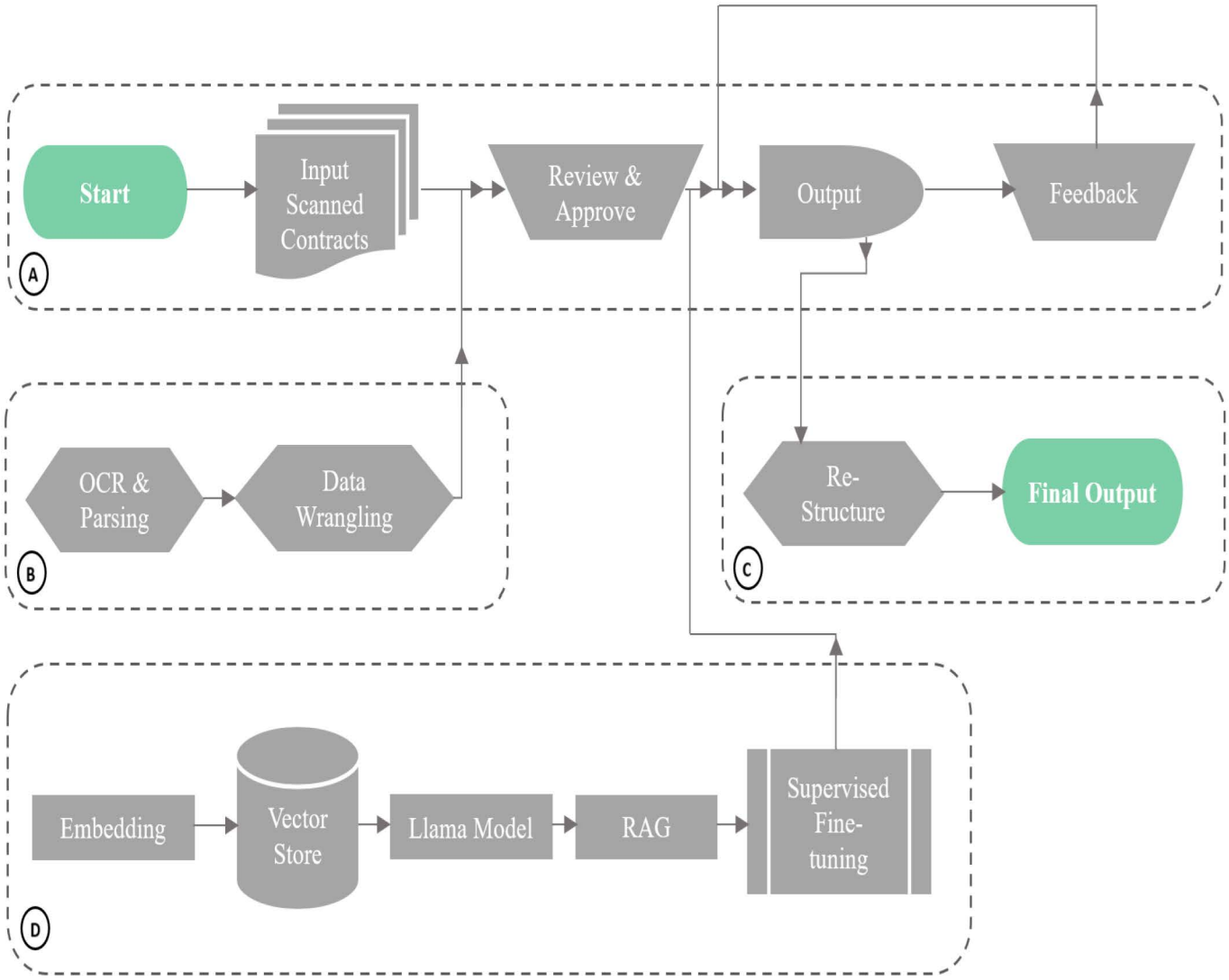

Fig. 1—Workflow

Dataset. In this workflow, the study was conducted on the technical documentations only of two expired contracts related to production and drilling operations. The study was limited to technical sections solely, which included the technical scope of work, remuneration tables, technical deviations and exceptions, and technical datasheets. The first contract covered a list of equipment, chemicals, services, and new technology, while the second contract primarily dealt with chemicals, tradenames, and some services. The documents of the first contract consisted of approximately 30 pages, while the second one had almost 50 pages. For the purpose of illustrating benchmarking capabilities only, the production contract was compared to two earlier contracts on the same subject to simulate a market scan. Due to privacy concerns, the data was masked and is not disclosed in this work, however, dummy examples were used to demonstrate the concepts.

OCR and Parsing. This dual approach was carried out simultaneously to ensure optimal data retrieval. OCR is a common natural language processing (NLP) technique, which converts text within images or scanned files into editable and searchable text formats making them effective with large unstructured documents (Dong et. al 2024). OCR consists of two steps, text detection and text recognition. However, the accuracy of OCR in scanning bid documents could be influenced by different factors such as poor resolution, the styling of characters in documents, the presence of hand-written signatures or comments, and the strikethrough text (Kissos and Dershowitz, 2016). The presence of such factors in bids makes the implementation of OCR solely unreliable for extracting key information from documents. Conversely,

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parsers are built on natural language processing to index information from scanned documents into structured, readable, searchable, and structured markdown format.

LlamaParse. In this workflow, LlamaParse was implemented as it uses a mixture of OCR along with a mixture of heuristic and machine learning models to reconstruct scanned documents. LlamaParse is a state-of-the-art table data extraction tool, which leverages natural language instructions to ensure good quality of data before it is passed to downstream LLMs. It is recognized as the world's first generative AI document parsing platform, which was built with LLMs, specifically for LLM applications (LlamaIndex).

LLM Frameworks. LlamaIndex and LangChain are two commonly utilized frameworks in the context of LLMs. Simply, the former offers a straightforward and adaptable interface between LLMs and pertinent external data, whereas the latter ensures the isolation between the developed LLM interface and the underlying LLM (Pinheiro et. al, 2023). LlamaIndex is an open-source library designed for context-augmented LLM applications, providing data connectors that could integrate with diverse data sources, data indices that systematically structure data, and a user interface that queries the indexed input to generate relevant outputs through different natural language engines. It comprises several core modules for indexing, storing, vectorizing, embedding, querying, retrieval, preprocessing, and synthesizing responses (LlamaIndex). Similarly, LangChain is an open-source library that facilitates the connection of LLMs to data sources and permits LLMs to interact with its environment. Essentially, it features modules for prompts, memory, chain, model, and agents (LangChain). Both frameworks are versatile and could be used for various downstream tasks such as QA tasks, semantic search, text summarization, and chatbot development.

LlamaIndex-LangChain. In this dual framework approach, LlamaIndex and LangChain are integrated to leverage their strengths. LangChain excels in applications that require complex interaction patterns and sustained context, which makes it ideal for chatbots, while LlamaIndex is preferred for tasks that demand precise document retrieval based on semantic relevance. The integration of both frameworks capitalizes on LlamaIndex's powerful indexing and retrieval capabilities and LangChain's high quality embeddings generation. This combination enables the development of efficient NLP applications. In particular, LlamaIndex benefits from the embeddings provided through LangChain, resulting in relevant retrievals when querying indexed data.

Embeddings. They are numerical representations of words, typically vectors, which encode information in a multi-dimensional space. This aids LLMs in comprehending complex documents data and identifying similarities more effectively by preserving sematic and syntactic relationships. Embedding models are pre-trained to transform information into dense vector forms. Common models include principal component analysis (PCA), singular value decomposition (SVD), Word2Vec, BERT, besides neural networks, such as sentence transformers, which are constructed based on transformers architecture, and excel at understanding semantics. Typically, a document is split, embedded, and then stored in a vector database, which stores data as high-dimensional vectors. Vector stores offer fast and accurate similarity search and retrieval for complex unstructured data, overcoming challenges of scalability bottlenecks, latency issues, and concurrency conflicts, which traditional databases might face (Han et al. 2023). Vector data storage includes techniques such as sharding, partitioning, caching, and replication, while search optimization may employ nearest neighbor search, approximate nearest neighbor search, or graph-based approaches. In addition, these data stores could support distributed learning, model compression, cross-modal search, reduced hallucination, and semantic search using specialized techniques such as ANN search, hashing, or quantization.

Vector Store. In this approach, allmpnet-base-v2 was implemented as the sentence transformer because of its effectiveness in capturing semantic similarity. It is an encoder model, which fine-tunes the pretrained Microsoft masked and permuted network (MPNet) base model. Unlike BERT, it captures the relationship among predicted tokens more effectively (Siino, 2023). For similarity searches, the dot product of embeddings – derived from sentence transformer and vector database – is used to relate texts to each other. In this context, an optimum chunk size of 1024 was chosen to split the document, noting that smaller chunk

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sizes could lead to more precise responses while larger ones could be more generic. To sum up, the allmpnet-base-v2 converted sentences into vectors in space where semantically similar sentences are close to each other. Then, those vectors are stored into a vector store index once the vector store is prepared. Then, LLM queries the database, transforming query into an embedding to fetch the most relevant response.

Large Language Models (LLMs). Built on the transformer architecture, Large Language Model Meta AI (LlaMa) leverages pre-normalization, SwiGLU activation function, and rotary embeddings. LlaMa is composed of three components: an embedding block, a stack of decoders, and a head block. Embedding layer is used to transform unstructured text into numerical vectors, known as tokens. Those tokens are subsequently directed to the decoder layer for further processing, where each decoder consists of multi-head self-attention (MSA) and a feed forward neural network (FFNN). Tokens are clustered in the MSA by attention maps obtained by the dot products between linear mappings of the input tokens, which are then processed by the FFNN (Han et al. 2024). LlaMa2 – implemented as the foundation model in this workflow – is an updated version of LlaMa1, architected with feed forward networks, attention layers, attention heads, vocabulary words, feature dimensions, hidden dimensions, and context-window size. LlaMa2 was released with 7B, 13B, and 70B parameters and was trained on 2 trillion tokens of data. It outperforms the existing open-source models and competes effectively with closed-source models (Touvron et al. 2023). It was trained using AdamW optimizer, with a tokenizer that employs a byte pair encoding (BPE) algorithm. Additionally, LlaMa2-chat was fine-tuned for dialogue use cases with a learning rate of 0.00002 and a sequence length of 4096 tokens. It also implements reinforcement learning human feedback (RLHF) to align the model response with human preferences.

LlaMa2. In this methodology, different iterations have been performed to optimize hyper-parameters. Ultimately, LlaMa2 context window was configured to 4096, defining the maximum number of tokens processed in a single input at a time. The limit for the generated tokens was set to 256, with the temperature set to zero to ensure that the most probable next token would always be predicted. LlaMa2 has sufficient context, so that token splitters are unnecessary, however, it is important to include overlaps when dividing the text into chunks to ensure that the relevant text would appear in two chunks.

Retrieval-Augmented Generation (RAG). Due to the tendency of LLMs to hallucinate, especially in domain-specific or knowledge-intensive tasks that go beyond their training data, retrieval-augmented generation (RAG) was developed. It enhances LLMs performance by ensuring the retrieval of relevant context from external source of knowledge, scanned documents in our case, through semantic similarity calculations. It involves two core systems: retrieval and generation. In retrieval phase, the naïve RAG employs the same encoding model utilized during the indexing phase. Then, it computes the similarity scores between the query vector and the vectors of indexed chunks. Next, the system retrieves the top K chunks that display the highest similarity scores, which are subsequently used as the context in prompt. In generation phase, the posed query and selected documents are synthesized into a coherent prompt to which LLM uses to formulate a response. Augmentation is implemented through iterative, recursive, or adaptive approaches, to enhance the efficiency of retrieval-generation process. However, naïve RAG encounters some drawbacks with retrieval challenges and generation difficulties, leading to potential hallucinations (Gao et al. 2023). Research could be divided into three mains stages, naïve, advanced, and modular RAGs, where the former adheres to the traditional retrieval and generation process, while the other two address shortcomings of the naïve module.

Advanced RAG. In this workflow, advanced RAG was implemented, focusing on enhancing retrieval quality by employing pre-retrieval and post-retrieval strategies and refining its indexing techniques through the use of a sliding window approach, fine-grained segmentation, and the incorporation of metadata. In addition, A dual approach of RAG and SFT was integrated, to achieve the optimum results, due to the fact that SFT could help minimize hallucinations, however, it could struggle with handling unfamiliar data.

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Supervised Fine Tuning (SFT). Fine tuning is more rigid, enabling extensive customization of model's behavior for specific tasks, as it involves making small adjustments to pre-trained models to achieve optimum results. However, fine tuning LLM parameters, in billions, is computationally expensive. Therefore, the parameter efficient fine-tuning (PEFT) was introduced, offering approaches to reduce the number of fine-tuned parameters (Bill and Eriksson, 2023). PEFT approaches could be categorized into additive, selective, reparametrized, and hybrid. Low-rank adaptation (LoRA) of LLMs is a common technique of reparametrized PEFT, which freezes weights of parameters, to minimize query and value layers in the attention metrics (Han et al. 2024).

Quantized Low Rank Adaptive (QLoRA). In this approach, QLoRA was implemented as it significantly reduces memory usage while preserving full 16-bit finetuning task performance. It uses a a high-preciison technique to quantize a pre-trained model to 4-bit without any performance degradation, then adds a small set of learnable LoRA weights that are tuned by backpropagating gradients through the quantized weights (Dettmers et al. 2024).

Reinforcement Learning from Human Feedback (RLHF). Inherently, pre-trained LLMs face the challenge of objective mismatch, which is the drift between training and deployment objectives as the goal of training is to capture language statistics while deployment aims to meeting user specified goals – procurement understanding in our case study. Using SFT solely replicates observable behaviors that the user finds desirable, without explicitly defining the criteria for this objective. Thereafter, employing a cognition-driven approach, which involves human feedback, was inevitable. It could be implemented through RLHF, which uses human feedback to guide the LLM learning process, so that the model could adapt to a particular task (Chaudhari et al. 2024).

Human Feedback. In this workflow, human feedback was implemented to evaluate the generated outputs and point out any mistakes made by the LLM. Additionally, human review and feedback were crucial during the parser output stage to fix issues, without any direct impact on LLM itself.

Chat Buffering. Chatbots are applications which simulate human interaction and conversations using NLP or LLMs to respond to user queries. Once the model was established successfully and generated reasonable outputs, it became inevitable to create a chatbot that stores interaction history. This allows the chatbot to maintain a longer context across multiple conversations, however, it is limited by the LLM context length. Basic concepts in chatbots include entity recognition, which is the extraction of parameter values from the input, and context, which is the string that stores source of data for the model. Chatbots could be classified based on knowledge domain into open domain and closed domain, or based on input processing as rule-based, retrieval-based, and generative-based (Adamopoulou and Moussiades, 2020).

Results

Data Transformation. The quality of data is the key for models performance as high-quality data provides clear insights for the model to work with and gains the trust of stakeholders. It also has a significant cost impact by improving efficiency and reducing the need for rework. In this first section of discussion, results of OCR and parsers data extraction from scanned documents are analyzed.

A comparison between LlamaParser and OCR tool (pytesseract) was demonstrated in (Fig. 2). The input section involved a sample of a scanned bidding document with dummy data and poor resolution – conditions that often resemble real-world data. The objective section assesses how effectively each method could transform data of scanned documents into structured data, ideally with the support from human feedback. The third section presents a comparison of the output from each technique. At this stage, it was observed that more than 25% of data were lost when OCR was employed solely, while less than 5% of data were reduced when using LlamaParser. However, even with LlamaParser, the accuracy was less than 90% upon human review. In the final section, human feedback and data analysis are suggested as corrective actions to improve output reliability. It is obvious that the best performing parser still made errors such as typos

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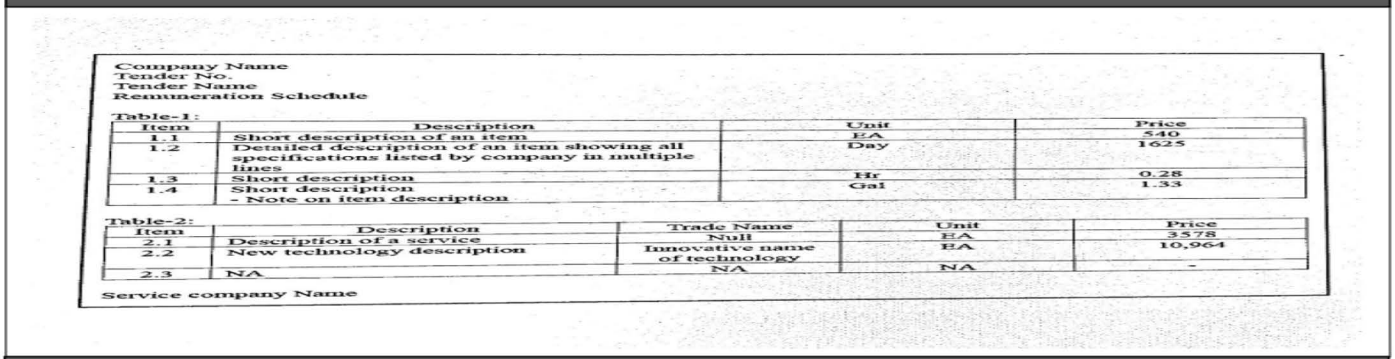
and data loss, particularly due to poor documents resolution and challenges in accurately interpreting multi-line or merged cells in tables.

Input:



Company Name
Tender No.
Tender Name
Remuneration Schedule

Table-1:

Item	Description	Unit	Price
1.1	Short description of an item	EA	540
1.2	Detailed description of an item showing all specifications listed by company in multiple lines	Day	1625
1.3	Short description	Hr	0.28
1.4	Short description - Note on item description	Gal	1.33

Table-2:

Item	Description	Trade Name	Unit	Price
2.1	Description of a service	Null	EA	3578
2.2	New technology description	Innovative name of technology	EA	10,964
2.3	NA	NA	NA	-

Service company Name

Objective:

Extracting data from a sample scanned documents using OCR only against using OCR with Parser
Next, showing the importance of Human-in-the-loop role at this early step for reviewing

OCR Only - Tesseract:

'Company Name\n\nTender No.\n\nTender Name\nRemuneration Schedule\n\nTable-i:\nItem Description\n\n1.1 Short description of an item.\n\n1.2 Detailed description of an item showing all\nspecifications listed by company in multiple\nlines\n\nShort description\n\nShort description\n\n- Note on item description\n\nDescription Trade Name\nNull\n\n21 Description of a service\n2.2 New technology description Innovative name\nof technology\n\nNA\n\n2.3 NA\n\nService company Name\n\n'

OCR + Parser:

'Company Name\n\n|Item|Description|Unit|Price|\n|---|---|---|---|\n|1.1|Short description of an item|EA|540|\n|1.2|Detailed description of an item showing all specifications listed by company in multiple lines|Day|1625|\n| | | |0.28|\n|1.3|Short description|Hr| |\n|1.44|Short description|Gal|1.33|\n\n# Table-2:\n\n|Item|Description|Trade Name|Unit|Price|\n|---|---|---|---|---|\n|2.1|Description of a service|Lunovative namle|EA|10,964|\n|2.2|New technology description of technology|NA|NA|NA|\n|2.3|Service company Name| | | |'

Corrective Action for the required Target:

'Company Name\n\n|Item|Description|Unit|Price|\n|---|---|---|---|\n|1.1|Short description of an item|EA|540|\n|1.2|Detailed description of an item showing all specifications listed by company in multiple lines|Day|1625|\n| | | |~~0.28~~|\n|1.3|Short description|Hr|0.28|\n|1.44|Short description|Gal|1.33|\n\n# Table-2:\n\n|Item|Description|Trade Name|Unit|Price|\n|---|---|---|---|---|\n|2.1|Description of a service|Null|EA|3578|\n|2.2|New technology description |~~Lu~~Innovative namle of technology |EA|10,964|\n|~~2.2|New technology description of technology~~2.3|NA|NA|NA|\n|~~2.3~~|Service company Name| | | |'

Fig. 2—The implementation of OCR and Parser to extract structured data from commercial tables in technical bids

As discussed earlier, OCR and parsing were the initial steps in the proposed workflow. Llamaindex parser was implemented effectively to extract unstructured data. It was followed by an automation and data analysis code along with human review to make sure that the extraction process was efficient, helping to eliminate errors associated with the extraction process. Ath this stage, errors encountered could be observed (Tables 2.1, 2.2, and 2.3). This step is considered mandatory, as it much improves the efficiency of LLMs. It should also be noted that having data in readable formats could have saved the redundant work of this step and could have eliminated the manual work required from users to review the extracted data manually, however, this is not usually the case in technical bids.

Table 2-1—Original parsed data after being transformed automatically into structured tables shows some errors with the extraction

Item	Description	Unit	Price	
1.1	Short description of an item	EA	540	
1.2	Detailed description of an item showing all specifications listed by company in multiple lines	Day	1625	
			0.28	

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Item	Description	Unit	Price	
1.3	Short description	Hr		
1.44	Short description	Gal	1.33	
Item	**Description**	**Trade Name**	**Unit**	**Price**
2.1	Description of a service	Lunovative namle	EA	10,964
2.2	New technology description of technology	NA	NA	NA
2.3	Service company Name			

Table 2-2—Application of data cleansing and automation code enhances structure of tables and correct errors resulted from inaccurate structuring.

Item	Description	Unit	Price	
1.1	Short description of an item	EA	540	
1.2	Detailed description of an item showing all specifications listed by company in multiple lines	Day	1625	
1.3	Short description	Hr	0.28	
1.4	Short description	Gal	1.33	
Item	**Description**	**Trade Name**	**Unit**	**Price**
2.1	Description of a service	Lunovative namle	EA	10,964
2.2	New technology description of technology	NA	NA	NA
2.3	Service company Name			

Table 2-3—Role of human-in-the-loop in reviewing the final tables, correcting typos, and adding missing data

Item	Description	Unit	Price	
1.1	Short description of an item	EA	540	
1.2	Detailed description of an item showing all specifications listed by company in multiple lines	Day	1625	
1.3	Short description	Hr	0.28	
1.4	Short description - Note on item description	Gal	1.33	
Item	**Description**	**Trade Name**	**Unit**	**Price**
2.1	Description of a service	Innovative namle	EA	3,578
2.2	New technology description	Innovative name of technology	EA	10,964
2.3	NA	NA	NA	NA

Scope of Work Evaluation. The primary objective was to compare the company's requirements in the technical table with descriptions and datasheets provided by the bidders. Performing this step manually often encounters challenges as bidders might claim that their equipment or material meet the requirements, yet the datasheets may not align with those claims.

Initially, named entity recognition (NER) was implemented to automatically extract the key specifications automatically from the technical description tables for each item. It was observed that the base model of BERT did not accurately extract specifications from technical tables, so that, a customized model was trained and optimized. Next, direct and fuzzy matching techniques were used to infer whether the required specifications were mentioned in the datasheets or not and to compare the numerical values of the

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requirements with the offered material datasheets. Results were provided in both Boolean and summary formats.

ROUGE-1 was initially intended to evaluate model performance quantitatively; however, it was observed that human oversight for evaluation was a necessity. Human feedback was used to evaluate the performance qualitatively, where it was observed that the model achieves accuracy of 0.83 on the used dataset.

Market Scan for Services. This approach is common in integrated oil companies when the validity of a contract is nearing expiration. The company conducts a benchmark study to compare the pricing of different service providers with other oil companies with an ultimate goal to decide whether to extend the validity of the existing contract with a considerable price increase from the service provider, or to start a new bidding process. A scope of work is developed, evaluation criteria are defined, and a table of commercial comparison is established to compare prices and decide accordingly. This process is often tedious and could be subject to bias or human error.

As mentioned in the dataset section in Methodology, the proposed workflow has been tested on different contracts to study the effectiveness of the introduced approach. The studied contracts included over 150 priced items, however, the established criteria for comparison were refined based on the expected operations in the following period of two years, narrowing the focus to 20 items. A table of criteria was established and input into the LLM, along with the other contracts. The output was the same table, populated with prices from the various contracts and any special notes about pricing notes.

Ablation Study without human feedback. In this study, the process began by enabling parsers to extract data from documents automatically, without including the step of human review and verification of the output. Then, the output was directly used with the LLM. It was observed that the workflow was executed successfully without any problems, however, the output failed to meet expectations. Upon investigation, it was found that issues stemmed from inaccurate parsing, leading to missing values and unstructured tables. The model achieved 0.70 accuracy on populating the data of the original contract, which should have been 1.0 as it represents the easiest targets for the model to query.

Ablation Study without supervised fine-tuning. In this scenario, human feedback was involved in the workflow to review the parsed values. Workflow started by querying the LLM to extract prices of each criteria item from the different contracts and compare them. It was found that the workflow achieved 1.0 accuracy on querying and populating data from the original contract, however, the workflow performance on the other contracts varied significantly to reach 0.85 and 0.65. Upon investigation, the root cause was found to be the variation in services description between contracts and the difference in pricing units.

Ablation Study with supervised fine-tuning. In this step, the purpose was to improve the model capability of similarity matching. Supervised fine tuning was carried out using examples from two of the contracts as training labeled data and testing it on the third contract. It was observed that the testing dataset showed 5% improvement when it came to issues related to similarity matching or variations in items description, however, no improvement was achieved when pricing units were different as it was not the focus of the proposed SFT.

Challenges and Future Work

The proposed workflow was limited to analyzing the technical sections of bid documents, and did not generalize about terms and conditions or other procurement aspects. Further contributions and additional development would be required to handle complex definitions and expressions commonly found in non-technical sections.

There was a discrepancy in the performance of the proposed workflow when evaluated across different contracts for market scan. While the introduced methodology excelled in semantic matching, it struggled with the variations in technical and remuneration tables structures, the contrast in units associated to each item, and the differences in pricing criteria per item – for example the same item could be priced based upon

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depth charge in one contract, and a lump-sum job in another. This encourages future work to establish criteria that could enable NLP to comprehend the foundational differences between evaluation and pricing criteria per contract, automate data transformations, and solicit user input for missing information and feedback. This shall ensure consistency across contracts before processing them with an LLM.

When comparing the proposed specifications with technical datasheets, it was found that evaluation based on human expertise and feedback were more effective than using ROUGE. Therefore, this research was limited to qualitative evaluation only, however, future studies should further explore ROUGE and other evaluation metrics that could assess the outputs effectively.

Running the proposed workflow locally was facilitated through the utilization of LlaMa and LlamaIndex, which are open-source tools, offering the option to store embeddings and operate LLMs on local machines. However, LlamaParse is not an open-source library, which limited this research to using samples of expired old documents only. Although LlamaParse claimed that the tool does not retain data, as it is permanently deleted after analysis, it is strongly recommended that future work explore alternative open-source libraries that compete with LlamaParse. This would enable the entire system to be deployed locally, addressing privacy concerns by eliminating the need for internet connectivity.

Conclusion

In this paper, a novel workflow was introduced to assist in technical bids evaluation in oil integrated companies. The study explored two scenarios: market scan surveys and the assessment of bidders' datasheets relevance to the company's proposed scope of work. LlaMa model was implemented as an open-source model, and the effectiveness was validated on two distinct contracts for drilling and production services. The findings suggest that LLMs exhibit significant potential in assisting technical engineers with the evaluation process, however, some notable drawbacks exist. In spite of this, it was concluded that incorporating human feedback could further enhance the process and maximize the output. This research shall be the groundwork for future studies on this topic, especially with the ongoing breakthrough in generative AI technologies.

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